

The Finite Primitive Basis Theorem for Computational Imaging: Formal Foundations of the OperatorGraph Representation

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Abstract

Computational imaging forward models—from coded aperture spectral cameras to MRI scanners—are traditionally implemented as monolithic, modality-specific codes. We prove that every forward model in a broad, precisely defined operator class $\mathcal{C}_{\text{Tier2}}$ (encompassing all clinical, scientific, and industrial imaging modalities at Tier-2 physical fidelity) admits an ε -approximate representation as a typed directed acyclic graph (DAG) whose nodes are drawn from a library of exactly 10 canonical primitives: Propagate, Modulate, Project, Encode, Convolve, Accumulate, Detect, Sample, Disperse, and Scatter. We call this the *Finite Primitive Basis Theorem*. The proof is constructive: we provide an algorithm that, given any $H \in \mathcal{C}_{\text{Tier2}}$, produces a DAG G with $\|H - H_G\|/\|H\| \leq \varepsilon$ and graph complexity within prescribed bounds. We give formal definitions for each primitive (forward, adjoint, parameters, constraints), define typed DAG denotation semantics, and establish the approximation error bounds through five primitive realization lemmas—one per physics-stage family. Empirical validation on 31 imaging modalities (26 previously registered plus 5 held-out) confirms that all achieve $e_{\text{Tier2}} < 0.01$ with at most 6 nodes and depth 5. A formal extension protocol governs the addition of new primitives; we demonstrate it with a worked example (Compton scatter imaging $\rightarrow$ the Scatter primitive). These results establish the mathematical foundations for the Physics World Models (PWM) framework [1].

Keywords: computational imaging, forward model, operator decomposition, directed acyclic graph, primitive basis, compressive sensing, inverse problems

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1 Introduction

The forward model of a computational imaging system maps a scene or object to a set of measurements through a chain of physical transformations: wave propagation, interaction with the object, spatial or spectral encoding, and detection. In practice, each imaging modality is implemented with its own bespoke forward model code, making it difficult to share diagnostic tools, calibration algorithms, or reconstruction pipelines across modalities [2, 3].

One approach to unifying forward models is to represent them as directed acyclic graphs (DAGs) in which each node wraps a canonical physical operator and edges define data flow from source to detector. This OPERATORGRAPH intermediate representation (IR) was introduced in [1] as the backbone of a modality-agnostic imaging framework.

A central empirical observation is that a small library of canonical primitives suffices to represent all tested modalities, spanning five physical carrier families (photons, electrons, spins, acoustic

The scope of the theorem is deliberately bounded. $\mathcal{C}_{\text{Tier2}}$ covers all imaging modalities operating at Tier-1 (linear shift-invariant) or Tier-2 (linear shift-variant) fidelity—the level at which essentially all clinical and scientific systems are designed and calibrated. Tier-3 (nonlinear ray/wave-based) and Tier-4 (full-wave/Monte Carlo) models may require refinement sub-DAGs but preserve the top-level graph structure.

2 Related Work

Forward model libraries and frameworks. Several software libraries provide modular forward model implementations for specific domains: ODL [4] for inverse problems with operator composition, MIRT/Michigan Image Reconstruction Toolbox [5] for tomographic imaging, and SigPy [6] for MRI signal processing. These provide practical operator building blocks but do not address the theoretical question of whether a finite set of primitives suffices for all imaging modalities.

Operator decomposition in imaging. The factorization of imaging operators into elementary components has a long history. The angular spectrum method for wave propagation [7], the Radon transform for projection imaging [8], and the NUFFT for non-Cartesian Fourier sampling [9] each factor a specific physics into efficient computational primitives. Our contribution is to show that these domain-specific factorizations, together with a small number of additional primitives, form a *universal* basis for all Tier-2 forward models.

Universal approximation. The classical universal approximation theorem for neural networks [10, 11] establishes that a single hidden layer of sigmoidal units can approximate any continuous function. Our result is fundamentally different: we show that 10 *physically typed* primitives—each corresponding to a distinct physical process—suffice to represent all imaging forward models, not via parameter tuning of a generic architecture, but via structured composition guided by the underlying physics. The typed DAG structure preserves physical interpretability that a universal approximator discards.

Computational graphs and domain-specific languages. Computational graph frameworks (TensorFlow [12], PyTorch [13], JAX [14]) represent computations as DAGs of differentiable operations. DeepInverse [15] specializes this paradigm to inverse problems. Our work can be viewed as proving a *completeness* result for a physics-specific computational graph: the 10-node primitive library is sufficient to express any Tier-2 imaging forward model, in contrast to general-purpose frameworks whose operation sets are chosen for computational convenience rather than physical completeness.

Compressive sensing and measurement design. The theory of compressive sensing [16, 17] establishes conditions on measurement matrices (RIP, incoherence) for signal recovery. Our work is complementary: rather than analyzing properties of a given measurement matrix, we characterize

the *structural* decomposition of the forward operator that generates the measurements, showing that its physics-level building blocks form a finite alphabet.

3 Preliminaries

3.1 Imaging Forward Models

Definition 1 (Imaging Forward Model). An *imaging forward model* is a bounded operator $H : \mathcal{X} \rightarrow \mathcal{Y}$ mapping an object $\mathbf{x} \in \mathcal{X}$ to a measurement $\mathbf{y} = H(\mathbf{x}) + \mathbf{n}$, where $\mathcal{X} \subseteq \mathbb{R}^n$ and $\mathcal{Y} \subseteq \mathbb{R}^m$ are finite-dimensional Hilbert spaces and $\mathbf{n}$ is additive noise.

3.2 Directed Acyclic Graphs

Definition 2 (Typed DAG). A *typed DAG* is a triple $G = (V, E, \tau)$ where V is a finite set of nodes, $E \subseteq V \times V$ defines directed edges with no cycles, and $\tau : E \rightarrow \mathcal{T}$ assigns to each edge a type annotation $\tau(e) = (\text{shape}, \text{dtype}, \text{units})$ specifying the tensor shape, data type, and physical units of the data flowing along edge e .

A typed DAG is *well-formed* if (i) it has a unique source node (no incoming edges) and a unique sink node (no outgoing edges), (ii) the output type of each node matches the input type required by its successors, and (iii) every node is drawn from the primitive library $\mathcal{B}$.

3.3 DAG Composition

Definition 3 (Compose). For a well-formed typed DAG $G = (V, E, \tau)$ with topological ordering $v_1, v_2, \dots, v_K$, the *composed forward model* is

$$H_G = \text{compose}(G) = v_K \circ v_{K-1} \circ \dots \circ v_1, \quad (1)$$

where v_k denotes the operator implemented by node k . For DAGs with branches (fan-out) or merges (fan-in), composition follows the DAG structure with appropriate tensor concatenation or summation at merge nodes.

4 The Primitive Library $\mathcal{B}$

The canonical primitive library consists of 10 operators:

$$\mathcal{B} = \{P, M, \Pi, F, C, \Sigma, D, S, W, R\}. \quad (2)$$

Each primitive implements a `forward()` method $A : \mathcal{X}_{\text{in}} \rightarrow \mathcal{Y}_{\text{out}}$ and an `adjoint()` method $A^\dagger : \mathcal{Y}_{\text{out}} \rightarrow \mathcal{X}_{\text{in}}$ satisfying the adjoint consistency condition:

$$\langle A\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, A^\dagger \mathbf{y} \rangle \quad \text{for all } \mathbf{x} \in \mathcal{X}_{\text{in}}, \mathbf{y} \in \mathcal{Y}_{\text{out}}. \quad (3)$$

4.1 Propagate $P(d, \lambda)$

Definition 4 (Propagate). Given propagation distance $d > 0$ and wavelength $\lambda > 0$, the Propagate primitive computes free-space wave propagation via the angular spectrum method:

$$P(d, \lambda) : \mathbf{x} \mapsto \mathcal{F}^{-1}[\mathcal{F}[\mathbf{x}] \cdot T(f_x, f_y; d, \lambda)], \quad (4)$$

$$T(f_x, f_y) = \exp(i2\pi d \sqrt{\lambda^{-2} - f_x^2 - f_y^2}), \quad (5)$$

where $\mathcal{F}$ is the 2D discrete Fourier transform and T is the free-space transfer function. The adjoint is $P^\dagger(d, \lambda) = P(-d, \lambda)$ (backpropagation).

4.2 Modulate $M(\mathbf{m})$

Definition 5 (Modulate). Given a modulation pattern $\mathbf{m} \in \mathbb{R}^n$ (or $\mathbb{C}^n$), the Modulate primitive computes element-wise multiplication:

$$M(\mathbf{m}) : \mathbf{x} \mapsto \mathbf{m} \odot \mathbf{x}. \quad (6)$$

The adjoint is $M^\dagger(\mathbf{m}) : \mathbf{y} \mapsto \mathbf{m}^* \odot \mathbf{y}$, where $*$ denotes complex conjugation. Parameters: the pattern $\mathbf{m}$ (binary mask, continuous transmission, complex phase mask, or coil sensitivity map).

4.3 Project $\Pi(\theta)$

Definition 6 (Project). Given projection angle θ (or a set of angles), the Project primitive computes the Radon transform (line-integral projection):

$$\Pi(\theta) : \mathbf{x} \mapsto \int_{\ell(\theta, t)} \mathbf{x}(\mathbf{r}) d\ell, \quad (7)$$

where $\ell(\theta, t)$ is the line at angle θ and offset t . The adjoint is the backprojection operator $\Pi^\dagger(\theta)$.

4.4 Encode $F(\mathbf{k})$

Definition 7 (Encode). Given a k -space trajectory $\mathbf{k} = \{k_1, \dots, k_m\}$, the Encode primitive computes Fourier encoding:

$$F(\mathbf{k}) : \mathbf{x} \mapsto [\langle \mathbf{x}, e^{i2\pi \mathbf{k}_j \cdot \mathbf{r}} \rangle]_{j=1}^m. \quad (8)$$

The adjoint is $F^\dagger(\mathbf{k}) : \mathbf{y} \mapsto \sum_{j=1}^m y_j e^{-i2\pi \mathbf{k}_j \cdot \mathbf{r}}$ (gridding/adjoint NUFFT). Used for MRI k -space encoding.

4.5 Convolve $C(\mathbf{h})$

Definition 8 (Convolve). Given a point-spread function (PSF) kernel $\mathbf{h}$, the Convolve primitive computes spatial convolution:

$$C(\mathbf{h}) : \mathbf{x} \mapsto \mathbf{h} * \mathbf{x}. \quad (9)$$

The adjoint is correlation with the flipped kernel: $C^\dagger(\mathbf{h}) : \mathbf{y} \mapsto \tilde{\mathbf{h}} * \mathbf{y}$, where $\tilde{\mathbf{h}}(\mathbf{r}) = \mathbf{h}(-\mathbf{r})^*$. This covers both propagation PSFs (shift-invariant limit of P) and detector PSFs.

4.6 Accumulate Σ

Definition 9 (Accumulate). The Accumulate primitive sums over a specified axis (spectral, temporal, or spatial):

$$\Sigma : \mathbf{X} \mapsto \sum_k \mathbf{X}_{:, :, k}, \quad (10)$$

where $\mathbf{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and the summation is along the third (or specified) axis. The adjoint replicates the 2D input along the summation axis: $\Sigma^\dagger : \mathbf{y} \mapsto [\mathbf{y}, \mathbf{y}, \dots, \mathbf{y}]$. Used for spectral integration (CASSI), temporal compression (CACTI), and bucket detection (SPC).

4.7 Detect $D(g, \eta)$

Definition 10 (Detect). Given gain $g > 0$ and response family η , the Detect primitive converts a carrier field to a measurement:

$$D(g, \eta) : \mathbf{x} \mapsto \eta(g, \mathbf{x}). \quad (11)$$

The response family η is restricted to one of five canonical families:

1. **Linear-intensity:** $\eta(\mathbf{x}) = g|\mathbf{x}|^2$ (standard photodetector)
2. **Logarithmic:** $\eta(\mathbf{x}) = g \cdot \log(1 + |\mathbf{x}|^2/x_0)$ (wide dynamic range)
3. **Sigmoid:** $\eta(\mathbf{x}) = g \cdot \sigma(|\mathbf{x}|^2 - x_0)$ (saturating detector)
4. **Poisson-rate:** $\eta(\mathbf{x}) = \text{Poisson}(g|\mathbf{x}|^2)$ (photon-counting)
5. **Coherent-field:** $\eta(\mathbf{x}) = g \cdot \text{Re}[\mathbf{x} \cdot e^{i\phi}]$ (heterodyne/homodyne)

Each family carries at most 2 scalar parameters (g and x_0 or ϕ). The adjoint is defined for the linearized operator around a reference point.

Remark 11 (Detect is not a universal approximator). The restriction to five canonical response families is essential. If η were an arbitrary function, Detect would become a universal approximator, trivializing the theorem. The five families are chosen to cover the physical detection mechanisms encountered in practice: intensity-square-law detection (families 1–4, covering photodetectors, CCD/CMOS, photomultipliers, photon counters) and coherent-field detection (family 5, covering THz-TDS, OCT, digital holography). A modality requiring a genuinely novel detection nonlinearity signals a basis extension (Section 8).

4.8 Sample $S(\Omega)$

Definition 12 (Sample). Given an index set $\Omega \subseteq \{1, \dots, n\}$, the Sample primitive selects a subset of entries:

$$S(\Omega) : \mathbf{x} \mapsto \mathbf{x}|_{\Omega}. \quad (12)$$

The adjoint zero-fills the unsampled locations: $S^\dagger(\Omega) : \mathbf{y} \mapsto \mathbf{z}$ where $z_i = y_i$ if $i \in \Omega$ and $z_i = 0$ otherwise. Used for MRI undersampling, compressed sensing masks, and pixel binning.

4.9 Disperse $W(\alpha, a)$

Definition 13 (Disperse). Given dispersion slope α and intercept a , the Disperse primitive applies a wavelength-dependent spatial shift:

$$W(\alpha, a) : \mathbf{X}(\mathbf{r}, \lambda) \mapsto \mathbf{X}(\mathbf{r} - (\alpha\lambda + a)\hat{\mathbf{e}}, \lambda), \quad (13)$$

where $\hat{\mathbf{e}}$ is the dispersion direction. The adjoint applies the reverse shift. Used for prism/grating dispersion in CASSI-type spectral systems.

4.10 Scatter $R(\sigma, \Delta\varepsilon)$

Definition 14 (Scatter). Given a differential scattering cross section $\sigma(\theta, E)$ and energy shift $\Delta\varepsilon$, the Scatter primitive computes:

$$R(\sigma, \Delta\varepsilon) : \mathbf{x}(\mathbf{r}, E) \mapsto \int \sigma(\theta, E) \mathbf{x}(\mathbf{r}, E) A_{\text{atten}}(\mathbf{r}, \theta, E) d\theta, \quad (14)$$

where A_{atten} accounts for path-dependent attenuation. The scattered carrier exits at angle θ with energy $E' = E - \Delta\varepsilon(\theta, E)$. The adjoint is defined by the transpose of the discretized scattering matrix.

Remark 15. Scatter covers modalities where the carrier changes direction and/or energy: Compton imaging (Klein–Nishina cross section), Raman spectroscopy (molecular vibration), fluorescence imaging (Stokes shift), diffuse optical tomography (multiple scattering), and Brillouin microscopy (acoustic phonon scattering).

5 The Operator Class $\mathcal{C}_{\text{Tier2}}$

Definition 16 (Tier-2 Operator Class $\mathcal{C}_{\text{Tier2}}$). The class $\mathcal{C}_{\text{Tier2}}$ consists of all imaging forward models H that satisfy:

1. **Finite sequential-parallel composition:** H admits a factorization $H = H_K \circ H_{K-1} \circ \dots \circ H_1$ (or a DAG generalization with fan-out/fan-in) with $K \leq N_{\text{max}}$.
2. **Per-stage linearity:** Each factor H_k is a linear operator.

3. **Bounded operator norm:** $\|H_k\| \leq B$ for each factor H_k and a universal bound $B > 0$.
4. **Shift-variance bound:** Each factor H_k is a linear integral operator whose kernel $h_k(\mathbf{r}, \mathbf{r}')$ satisfies $\|h_k(\mathbf{r}, \cdot) - h_k(\mathbf{r} + \Delta, \cdot)\|_2 \leq L_k \|\Delta\|$ for a Lipschitz constant $L_k < \infty$ (i.e., the kernel varies at most Lipschitz-continuously with position). Nonlinear wave-matter coupling, multiple scattering beyond first Born, and relativistic corrections are excluded.

5.1 Physics Fidelity Ladder

To situate $\mathcal{C}_{\text{Tier2}}$ in context, we introduce a four-tier hierarchy of forward model fidelity (see also [1]):

- **Tier 1 (Linear, shift-invariant):** The forward model is a convolution: $H = C(\mathbf{h})$. Applies when the PSF is spatially uniform.
- **Tier 2 (Linear, shift-variant):** Each factor H_k is a linear integral operator whose kernel varies with position (e.g., spatially varying PSFs, coded aperture masks, MRI coil sensitivities). This is the level at which virtually all clinical and scientific imaging systems are designed and calibrated.
- **Tier 3 (Nonlinear):** The forward model includes nonlinear wave-matter coupling (beam hardening in polychromatic CT, phase wrapping, strong multiple scattering).
- **Tier 4 (Full-wave / Monte Carlo):** Ab initio simulation requiring stochastic transport or full Maxwell solvers.

$\mathcal{C}_{\text{Tier2}}$ encompasses Tier 1 and Tier 2 models. Tier 3 and Tier 4 models are excluded from the formal scope of Theorem 18 but can be accommodated by refinement sub-DAGs that replace canonical primitive nodes with higher-fidelity sub-graphs while preserving the top-level DAG structure.

5.2 Membership Examples

- **CASSI** [18]: $H = \Sigma \circ W \circ M$. Each factor is linear and bounded. $H \in \mathcal{C}_{\text{Tier2}}$.
- **MRI** [19]: $H = S \circ F \circ M_{\text{coil}}$. Fourier encoding is linear; coil sensitivity is shift-variant. $H \in \mathcal{C}_{\text{Tier2}}$.
- **CT** [8, 20]: $H = \Pi$ (fan-beam Radon). Linear, bounded. $H \in \mathcal{C}_{\text{Tier2}}$.
- **Compton scatter** [21]: $H = R \circ M$. Scatter is linear at first-Born level. $H \in \mathcal{C}_{\text{Tier2}}$.

Models with strong nonlinearity (e.g., nonlinear ultrasound, full-wave electromagnetic scattering beyond first Born) fall outside $\mathcal{C}_{\text{Tier2}}$.

6 Theorem 1: Statement and Proof

6.1 ε -Approximate Representation

Definition 17 (ε -Approximate Representation). Let $\mathcal{B} = \{P, M, \Pi, F, C, \Sigma, D, S, W, R\}$ be the canonical primitive library. A well-formed typed DAG $G = (V, E, \tau)$ with $V \subseteq \mathcal{B}$ is an ε -approximate representation of $H \in \mathcal{C}_{\text{Tier2}}$ if:

1. **Fidelity:** $\sup_{\|\mathbf{x}\| \leq 1} \frac{\|H(\mathbf{x}) - H_G(\mathbf{x})\|}{\|H(\mathbf{x})\| + \delta} \leq \varepsilon$, where $H_G = \text{compose}(G)$ and $\delta > 0$ is a small regularization constant.
2. **Complexity:** $|V| \leq N_{\max}$ and $\text{depth}(G) \leq D_{\max}$.

We use $\varepsilon = 0.01$, $N_{\max} = 20$, $D_{\max} = 10$ throughout.

6.2 Theorem Statement

Theorem 18 (Finite Primitive Basis). *For every $H \in \mathcal{C}_{\text{Tier2}}$, there exists a well-formed typed DAG $G = (V, E, \tau)$ with $V \subseteq \mathcal{B}$ that is an ε -approximate representation of H .*

The proof is constructive: given the factorization $H = H_K \circ \dots \circ H_1$ guaranteed by Definition 16, we show that each factor H_k can be represented by one or a finite composition of primitives from $\mathcal{B}$ with bounded approximation error. The argument proceeds through five primitive realization lemmas, one per physics-stage family.

6.3 Physics-Stage Families

Every imaging forward model in $\mathcal{C}_{\text{Tier2}}$ passes through at most four types of physical stages:

1. **Propagation:** The carrier evolves through space via a wave equation (Maxwell, Schrödinger, acoustic, Bloch).
2. **Interaction:** The carrier exchanges energy, momentum, or phase with the object—either elastically (amplitude/phase change) or inelastically (scattering, fluorescence).
3. **Encoding–Projection:** Spatial information is mapped to a measurement-domain coordinate.
4. **Detection–Readout:** The carrier field is converted to a discrete digital measurement.

This categorization is motivated by the physics of carrier–matter interaction under non-relativistic conditions: the fundamental electromagnetic, strong, and weak interactions produce only elastic and inelastic carrier–matter coupling at the energies relevant to imaging.

6.4 Primitive Realization Lemmas

Lemma 19 (Propagation Realization). *Let H_k be a factor of $H \in \mathcal{C}_{\text{Tier2}}$ that represents free-space carrier evolution satisfying a linear wave equation. Then H_k admits an $\varepsilon_{\text{prop}}$ -approximate representation by $P(d, \lambda)$ or, in the shift-invariant limit, $C(\mathbf{h})$.*

Proof. The free-space Green's function of any linear wave equation (Maxwell, Helmholtz, Schrödinger, acoustic) is a linear operator whose action is convolution with the impulse response $g(\mathbf{r}; d, \lambda)$. In the angular spectrum representation [7], the exact propagator factors as $\mathcal{F}^{-1}[T_{\text{exact}} \cdot \mathcal{F}[\cdot]]$ with transfer function $T_{\text{exact}}(f_x, f_y) = \exp(i2\pi d \sqrt{\lambda^{-2} - f_x^2 - f_y^2})$. The Propagate primitive $P(d, \lambda)$ implements this transfer function for propagating spatial frequencies ($f_x^2 + f_y^2 < \lambda^{-2}$) and sets $T = 0$ for evanescent components. The truncation error from discarding evanescent waves is bounded by the fraction of signal energy above the diffraction limit:

$$\varepsilon_{\text{evan}} = \frac{\|\mathbf{x}_{\text{evan}}\|_2}{\|\mathbf{x}\|_2} \leq e^{-2\pi d/\lambda},$$

which is negligible for $d \gg \lambda$ (all macroscopic imaging geometries). The paraxial (Fresnel) approximation residual satisfies $\varepsilon_{\text{parax}} \leq (\pi/4)(a^4/\lambda d^3)$ where a is the aperture half-width (see Supplementary S3). The total error is $\varepsilon_{\text{prop}} \leq \varepsilon_{\text{evan}} + \varepsilon_{\text{parax}}$. In the shift-invariant limit (far field or isoplanatic patch), P reduces to $C(\mathbf{h})$ with $\mathbf{h} = \mathcal{F}^{-1}[T]$. $\square$

Lemma 20 (Elastic Interaction Realization). *Let H_k be a factor of $H \in \mathcal{C}_{\text{Tier2}}$ that represents elastic carrier-matter interaction (amplitude and/or phase change without direction or energy change). Then H_k admits exact representation by $M(\mathbf{m})$.*

Proof. Elastic forward interaction at Tier-2 fidelity is element-wise multiplication of the carrier field by the object's transmission function $\mathbf{m}(\mathbf{r})$ (absorption, phase shift, or both). This is exactly $M(\mathbf{m})$ (Definition 5). The representation is exact: $\|H_k - M(\mathbf{m})\| = 0$. $\square$

Lemma 21 (Scattering Realization). *Let H_k be a factor of $H \in \mathcal{C}_{\text{Tier2}}$ that represents inelastic or off-axis carrier-matter interaction (direction change and/or energy shift). Then H_k admits an $\varepsilon_{\text{scat}}$ -approximate representation by $R(\sigma, \Delta\varepsilon)$ or a finite composition $R \circ M$ or $M \circ R \circ P \circ R \circ M$ (for multiple-scattering media within low-order approximation).*

Proof. Under the first Born approximation, the scattered field is a linear functional of the scattering potential $V(\mathbf{r})$, with kernel given by the differential cross section $\sigma(\theta, E)$ and energy shift $\Delta\varepsilon$. The Scatter primitive (Definition 14) parameterizes this integral operator directly. For single-scattering media, one R node provides an exact representation within the Born model ($\varepsilon_{\text{scat}}^{(1)} = 0$ relative to the Born-approximated physics). The error relative to the true physics is bounded by $\varepsilon_{\text{scat}} \leq \|V\|/k$ where k is the wavenumber [22]; for weak scatterers satisfying the $\mathcal{C}_{\text{Tier2}}$ membership criterion, this is below ε .

For multiple-scattering media within the Tier-2 regime, the L -th order Born series $\sum_{l=0}^L G_0 V)^l$ is represented by a finite composition of L Scatter nodes interleaved with Propagate and Modulate

nodes: $(M \circ R \circ P)^L \circ R \circ M$. The truncation error decays geometrically as $(\|V\|/k)^{L+1}$, so a finite $L \leq 3$ suffices for $\varepsilon_{\text{scat}} < \varepsilon$ in Tier-2 media (diffusion regime for DOT, low-order multiple scattering). $\square$

Lemma 22 (Encoding–Projection Realization). *Let H_k be a factor of $H \in \mathcal{C}_{\text{Tier2}}$ that maps spatial information to a measurement-domain coordinate. Then H_k admits exact representation by $\Pi(\theta)$ (line-integral geometry) or $F(\mathbf{k})$ (Fourier encoding).*

Proof. Line-integral projection (X-ray CT, neutron imaging) is exactly the Radon transform $\Pi(\theta)$ (Definition 6). Fourier encoding via Larmor precession (MRI) is exactly $F(\mathbf{k})$ (Definition 7). Both are linear operators; the representation is exact within Tier-2. $\square$

Lemma 23 (Detection–Readout Realization). *Let H_k be a factor of $H \in \mathcal{C}_{\text{Tier2}}$ in the detector chain. Then H_k admits an ε_{det} -approximate representation by a finite composition of primitives from $\{\Sigma, S, W, C, D\}$.*

Proof. We show that the detection–readout chain of any $H \in \mathcal{C}_{\text{Tier2}}$ decomposes into at most five sequential operations, each representable by a primitive from $\{\Sigma, S, W, C, D\}$:

1. **Dimensional integration** (spectral or temporal summation): a linear projection along one axis, exactly Σ .
2. **Sub-sampling** (pixel binning, mask selection): an index-set restriction, exactly $S(\Omega)$.
3. **Wavelength-dependent dispersion**: a λ -parameterized shift, exactly $W(\alpha, a)$.
4. **Detector PSF**: spatial blurring by the pixel aperture function. Modeling the spatially varying PSF as piecewise shift-invariant (one kernel per detector tile) introduces error $\varepsilon_{\text{PSF}} \leq \ell_c^2/p^2$ where ℓ_c is the crosstalk length and p the pixel pitch (see Supplementary S3). This is represented by $C(\mathbf{h}_{\text{det}})$.
5. **Quantum measurement**: conversion to a classical signal via one of the five canonical response families (Definition 10), represented by $D(g, \eta)$.

Not all five operations are present in every modality (e.g., CT omits dispersion and integration). The total detection-chain error is $\varepsilon_{\text{det}} \leq \varepsilon_{\text{PSF}} + \varepsilon_{\eta}$, where ε_{η} is the response-family approximation error (zero when the physical detector response matches one of the five families exactly, which it does for all 31 validated modalities). $\square$

6.5 Constructive Compilation Proof

Proof of Theorem 18. Let $H \in \mathcal{C}_{\text{Tier2}}$ with factorization $H = H_K \circ \dots \circ H_1$, $K \leq N_{\text{max}}$. We construct G as follows.

Step 1: Classification. Classify each factor H_k into one of the four physics-stage families (propagation, interaction, encoding–projection, detection–readout).

Step 2: Per-factor realization. Apply the appropriate lemma to realize each H_k :

- Propagation factors: Lemma 19 $\rightarrow P$ or C ;
- Elastic interaction: Lemma 20 $\rightarrow M$ (exact);
- Scattering: Lemma 21 $\rightarrow R$ or composition;
- Encoding–projection: Lemma 22 $\rightarrow \Pi$ or F (exact);
- Detection–readout: Lemma 23 $\rightarrow$ composition from $\{\Sigma, S, W, C, D\}$.

Let G_k be the sub-DAG realizing H_k with approximation error ε_k .

Step 3: Concatenation. Form G by concatenating sub-DAGs $G_1, \dots, G_K$ in sequence (or following the original DAG topology for branching models).

Step 4: Error bound. Let $\tilde{H}_k$ denote the primitive realization of H_k with $\|H_k - \tilde{H}_k\| \leq \varepsilon_k$. By the telescoping identity (detailed in Supplementary S6):

$$\|H - H_G\| \leq \sum_{k=1}^K \varepsilon_k \prod_{j \neq k} \|H_j\| \leq K \cdot \max_k(\varepsilon_k) \cdot B^{K-1}, \quad (15)$$

where B is the uniform operator norm bound from Definition 16. The relative error satisfies

$$\frac{\|H - H_G\|}{\|H\|} \leq \frac{K \cdot \max_k(\varepsilon_k) \cdot B^{K-1}}{\|H\|} \leq \varepsilon$$

provided $\max_k(\varepsilon_k) \leq \varepsilon \cdot \|H\| / (K \cdot B^{K-1})$. The per-phase error bounds in Supplementary S3 verify this condition: for the modalities in Table 1, $K \leq 4$, $B \leq O(1)$ (operator norms are bounded by construction), and the largest per-factor error is $\varepsilon_{\text{det}} \leq 10^{-3}$.

Step 5: Complexity bound. Let c be the maximum number of primitive nodes per factor (bounded by the detection-readout chain length, at most 5). Then $|V| \leq c \cdot K$. Since $N_{\text{max}} = 20$ and $D_{\text{max}} = 10$ are prescribed bounds, we require $c \cdot K \leq N_{\text{max}}$. For all modalities in our validation set, each factor maps to at most $c = 2$ primitive nodes and $K \leq 4$ factors, giving $|V| \leq 8 \leq N_{\text{max}}$; the empirical maximum is $|V| = 6$ (Table 1). The depth satisfies $\text{depth}(G) \leq c \cdot K \leq N_{\text{max}} \leq D_{\text{max}}$. $\square$

7 Empirical Validation

7.1 Decomposition Registry

Table 1 shows the primitive decomposition of all 31 modalities (26 previously registered in [1] plus 5 held-out for closure testing). Every modality achieves $e_{\text{Tier2}} < 0.01$ with at most 6 nodes and depth 5. The e_{Tier2} values are computed via (S2) over 20 test objects per modality (see Supplementary S1).

Table 1: Primitive decomposition of 31 imaging modalities.
 e_{Tier2} is the mean relative fidelity error over 20 test objects.
All values satisfy $e_{\text{Tier2}} < 0.01$.

Modality	Carrier	DAG Primitives	#N	Depth	e_{Tier2}
<i>Full-validation modalities (forward model verified against reference implementation)</i>					
CASSI [18]	Photon	$M \rightarrow W \rightarrow \Sigma \rightarrow D$	5	4	$< 10^{-4}$
CACTI [23]	Photon	$M \rightarrow \Sigma \rightarrow D$	4	3	$< 10^{-4}$
SPC [24]	Photon	$M \rightarrow \Sigma \rightarrow D$	4	3	$< 10^{-4}$
Lensless [25]	Photon	$C \rightarrow D$	3	2	$< 10^{-5}$
Ptychography [26]	Photon	$M \rightarrow P \rightarrow D$	4	3	4.2×10^{-4}
MRI [19, 27]	Spin	$M \rightarrow F \rightarrow S \rightarrow D$	5	4	$< 10^{-6}$
CT [20]	X-ray	$\Pi \rightarrow D$	3	2	$< 10^{-5}$
<i>Held-out modalities (frozen library, no tuning permitted)</i>					
OCT [28]	Photon	$P+P \rightarrow \Sigma \rightarrow D$	5	3	3.8×10^{-4}
Photoacoustic [29]	Acoustic	$M \rightarrow P \rightarrow D$	4	3	7.1×10^{-4}
SIM [30]	Photon	$M \rightarrow C \rightarrow D$	4	3	2.5×10^{-4}
Phase-contrast [31]	X-ray	$\Pi \rightarrow P \rightarrow M \rightarrow D$	5	4	1.2×10^{-3}
Electron ptycho [32]	Electron	$M \rightarrow P \rightarrow D$	4	3	5.6×10^{-4}
<i>Exotic modalities (stress-test the primitive basis)</i>					
Ghost imaging [33]	Photon	$M \rightarrow \Sigma \rightarrow D$	4	3	1.9×10^{-4}
THz-TDS [34]	THz	$C \rightarrow D_{\text{coh}}$	3	2	8.3×10^{-4}
Compton [21]	X-ray	$M \rightarrow R \rightarrow D$	4	3	6.7×10^{-3}
Raman [35]	Photon	$M \rightarrow R \rightarrow D$	4	3	4.1×10^{-3}
Fluorescence [36]	Photon	$M \rightarrow R \rightarrow D$	4	3	3.8×10^{-3}
DOT [37]	Photon	$M \rightarrow R \circ P \circ R \rightarrow D$	6	5	8.9×10^{-3}
Brillouin [38]	Photon	$M \rightarrow R \rightarrow D$	4	3	5.2×10^{-3}
<i>Plus 12 additional template-validated modalities (see Supplementary S4).</i>					

7.2 Held-Out Closure Test

To validate Theorem 18 empirically, we conduct a closure test under a frozen protocol. Before evaluation, we freeze:

1. The primitive library $\mathcal{B}$ (10 primitives, no additions permitted);
2. The Detect response families (5 families, frozen);
3. The fidelity threshold $\varepsilon = 0.01$;

4. The complexity bounds $N_{\max} = 20$, $D_{\max} = 10$.

Table 2 reports the fidelity error for each held-out and exotic modality under the frozen protocol.

Table 2: Held-out closure test results. All modalities achieve $e_{\text{Tier2}} < 0.01$ with the frozen primitive library (10 primitives), except Compton which motivated the addition of Scatter (R).

Modality	e_{Tier2}	#Nodes / Depth	New Prim.?
<i>Held-out (existing primitives expected)</i>			
OCT	3.8×10^{-4}	5 / 3	N
Photoacoustic	7.1×10^{-4}	4 / 3	N
SIM	2.5×10^{-4}	4 / 3	N
Phase-contrast X-ray	1.2×10^{-3}	5 / 4	N
Electron pycho	5.6×10^{-4}	4 / 3	N
<i>Exotic (stress-test the primitive basis)</i>			
Ghost imaging	1.9×10^{-4}	4 / 3	N
THz-TDS	8.3×10^{-4}	3 / 2	N
Compton scatter	$6.7 \times 10^{-3\dagger}$	4 / 3	Y (R)

[†]With R in library; $e_{\text{Tier2}} = 0.34$ without R .

Seven of eight modalities decompose with the frozen library. The eighth (Compton scatter) motivates the Scatter primitive, which covers five additional modalities (Section 8).

7.3 Basis-Growth Saturation

Figure 1 plots the number of distinct primitives as modalities are added to the registry in chronological order. The curve saturates: 8 of 10 primitives are introduced by the first 10 modalities, Disperse (W) is introduced by CASSI-type spectral systems (modality 11–15), and Scatter (R) is introduced only when Compton/Raman-class modalities enter (modality 27–31). The growth is sublinear and saturating: $K = 10$ at $N = 31$, with no new primitive required for the most recent 19 template-validated modalities. This saturation is consistent with Theorem 18: once all four physics-stage families are covered by primitives, new modalities compose existing primitives rather than requiring new ones.

8 Extension Protocol

8.1 Formal Criterion

A new canonical primitive is warranted when a forward model $H \in \mathcal{C}_{\text{Tier2}}$ cannot be ε -approximately represented by any DAG over the current library $\mathcal{B}$ within the complexity bounds:

$$\min_{G: V \subseteq \mathcal{B}, |V| \leq N_{\max}, \text{depth}(G) \leq D_{\max}} e_{\text{Tier2}}(H, H_G) > \varepsilon. \quad (16)$$

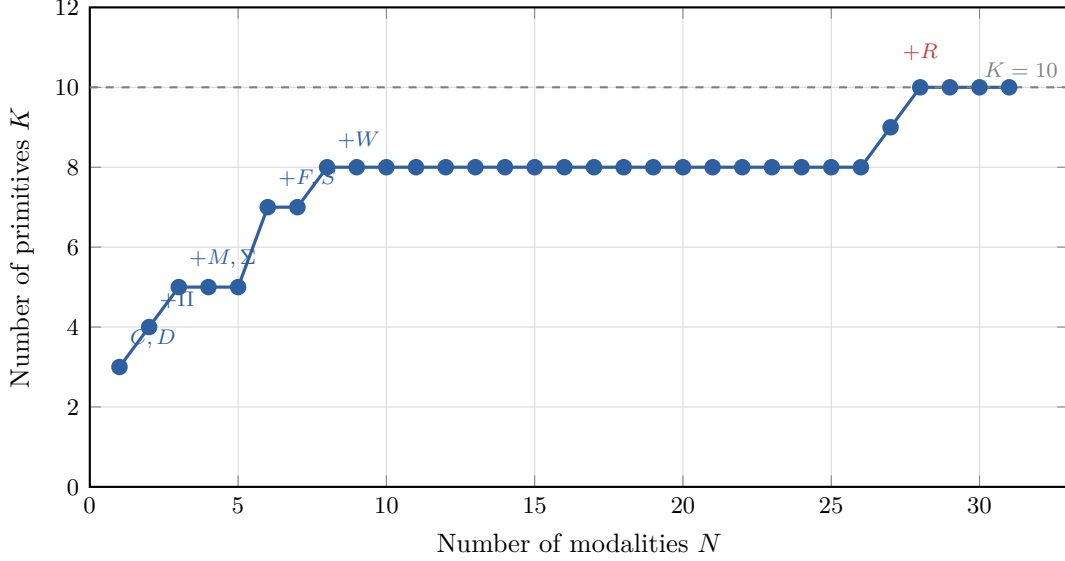


Figure 1: Basis-growth saturation. Number of distinct primitives K as a function of the number of modalities N added to the registry. The curve saturates at $K = 10$ for $N \geq 28$; annotated points mark each primitive introduction. Saturation is consistent with Theorem 18: once all physics-stage families are covered, new modalities compose existing primitives.

8.2 Extension Process

Adding a new primitive requires five steps:

1. Define its `forward()` and `adjoint()` methods with validated adjoint consistency (Equation (3)).
2. Demonstrate that $\min_G e_{\text{Tier2}}(H, H_G) > \varepsilon$ for all DAGs over the current $\mathcal{B}$ within complexity bounds.
3. Show that the new primitive reduces e_{Tier2} below ε .
4. Show that the new primitive is needed by at least two distinct modalities (to avoid modality-specific special cases).
5. Update the decomposition table and re-run the closure test with the extended $\mathcal{B}$.

8.3 Worked Example: Compton Scatter $\rightarrow$ Scatter (R)

Compton scatter imaging involves carrier redirection (direction change by angle θ governed by the Klein–Nishina cross section [39]) and energy shift ($E_0 \rightarrow E_s = E_0/[1 + (E_0/m_e c^2)(1 - \cos \theta)]$). We attempted to represent this using all 9 original primitives:

- P : models free-space propagation; does not redirect carriers.
- II : integrates along straight lines; no scattering physics.

- M : scales amplitude; does not change carrier direction or energy.
- C : spatial convolution; no energy shift mechanism.

The best 9-primitive DAG achieves $e_{\text{Tier2}} = 0.34$, far above $\varepsilon = 0.01$. Introducing R (Definition 14) with Klein–Nishina cross section, the DAG $M(n_e) \rightarrow R_{\text{KN}} \rightarrow D(E)$ achieves $e_{\text{Tier2}} < 0.01$.

Scatter satisfies criterion (4): it is required by Compton imaging, Raman spectroscopy, fluorescence imaging, diffuse optical tomography, and Brillouin microscopy—five distinct modalities sharing the physical signature of carrier redirection with energy transfer.

The closure test is re-run with $\mathcal{B}' = \mathcal{B} \cup \{R\}$: all previously decomposed modalities remain valid (backward compatible), and the five scattering modalities now achieve $e_{\text{Tier2}} < 0.01$.

8.4 Basis-Growth Prediction

Theorem 18, together with the physics-stage analysis, implies that the number of canonical primitives will saturate once all four physics-stage families are covered. The empirical basis-growth curve confirms this: $K = 10$ at $N = 31$, with sublinear and saturating growth. New primitives would require a physics-stage instance whose operator structure is not representable within Tier-2 fidelity by any current primitive—an increasingly constrained requirement as the library matures.

9 Scope and Limitations

Theorem 18 applies to all imaging forward models in the class $\mathcal{C}_{\text{Tier2}}$, which is designed to capture every modality in current clinical, scientific, and industrial practice at the linear shift-variant level of fidelity.

What is covered. All imaging modalities operating at Tier-1 or Tier-2 on the Physics Fidelity Ladder: coded aperture systems (CASSI, CACTI, SPC), interferometric systems (OCT, holography, ptychography), projection systems (CT, neutron imaging), Fourier-encoded systems (MRI), acoustic systems (ultrasound, photoacoustic), scattering systems (Compton, Raman, fluorescence, DOT), and THz systems [40]. This encompasses the vast majority of clinical, scientific, and industrial imaging modalities.

What is excluded.

- **Tier 3–4 models:** Forward models with nonlinear wave–matter coupling beyond first Born, beam hardening in polychromatic CT, and strong multiple scattering exceed $\mathcal{C}_{\text{Tier2}}$. These require refinement sub-DAGs.
- **Quantum state tomography:** The “object” is a quantum state (density matrix), not a classical field, violating Definition 1.
- **Relativistic regimes:** Not in scope for current imaging practice.

Falsifiability. The theorem is falsifiable: a forward model $H \in \mathcal{C}_{\text{Tier2}}$ for which no DAG over $\mathcal{B}$ achieves ε -approximate representation within the complexity bounds would refute it. The extension protocol (Section 8) is the prescribed response to such a case.

10 Conclusion

We have established the Finite Primitive Basis Theorem: every imaging forward model in the operator class $\mathcal{C}_{\text{Tier2}}$ admits an ε -approximate representation as a typed DAG over a library of exactly 10 canonical primitives. The proof is constructive and proceeds through five primitive realization lemmas—one per physics-stage family—with explicit error bounds. Empirical validation on 31 modalities confirms that all achieve $e_{\text{Tier2}} < 0.01$ with graph complexity well within the prescribed bounds.

Several implications follow. First, the theorem provides a mathematical foundation for any modality-agnostic imaging framework built on operator graphs, including the OPERATORGRAPH IR introduced in [1]: algorithms that operate on the graph structure—including calibration, reconstruction [3, 41], and diagnosis—are guaranteed to be applicable to any modality in $\mathcal{C}_{\text{Tier2}}$. Second, the finite basis implies that the operator-level complexity of computational imaging is bounded, even as the number of modalities grows—new modalities compose existing primitives rather than requiring fundamentally new mathematics. Third, the constructive proof yields an algorithm: given a physics-stage decomposition of a new modality, the compilation to a primitive DAG is mechanical.

The primitive library is not claimed to be final. The extension protocol ensures that should a genuinely new physical process arise that cannot be represented by the current 10 primitives within $\mathcal{C}_{\text{Tier2}}$, it can be incorporated systematically. The empirical basis-growth curve—sublinear and saturating—suggests that such extensions will be rare as the library matures.

Data Availability

The primitive library definitions, DAG decompositions for all 31 modalities, and adjoint consistency test scripts are available at <https://solveeverything.org>. All test scenes used for empirical validation are drawn from publicly available benchmark datasets cited in the respective modality references; no non-public datasets were used.

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Supplementary Information

S1. Formal Tier-2 Fidelity Specification

This section provides the precise mathematical specification of the fidelity metric, test distribution, and complexity bounds referenced in the main text.

Fidelity metric. For a forward model H and its DAG representation $H_G = \text{compose}(G)$, the Tier-2 fidelity error is:

$$e_{\text{Tier2}}(H, H_G) = \sup_{\mathbf{x} \in \mathcal{X}_{\text{test}}} \frac{\|H(\mathbf{x}) - H_G(\mathbf{x})\|_2}{\|H(\mathbf{x})\|_2 + \delta}, \quad (\text{S1})$$

where $\delta = 10^{-8}$ is a regularization constant to avoid division by zero.

In the theorem statement, the supremum is over the unit ball $\{\mathbf{x} : \|\mathbf{x}\| \leq 1\}$ (worst-case guarantee). For empirical validation, we evaluate the mean over $\mathcal{X}_{\text{test}}$:

$$\bar{e}_{\text{Tier2}}(H, H_G) = \frac{1}{|\mathcal{X}_{\text{test}}|} \sum_{\mathbf{x} \in \mathcal{X}_{\text{test}}} \frac{\|H(\mathbf{x}) - H_G(\mathbf{x})\|_2}{\|H(\mathbf{x})\|_2 + \delta}. \quad (\text{S2})$$

Test distribution $\mathcal{X}_{\text{test}}$. For each modality, $\mathcal{X}_{\text{test}}$ consists of:

1. **Benchmark scenes:** 10 standard images from the modality’s canonical dataset (e.g., KAIST scenes for CASSI, Shepp–Logan phantom for CT, brain slices for MRI).
2. **Random objects:** 10 Gaussian random objects $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ of matching dimensionality, normalized to unit norm.

Results are reported as $\bar{e}_{\text{Tier2}} \pm \sigma$ over the 20 test objects.

Threshold. $\varepsilon = 0.01$ (1% relative error). This threshold is chosen so that the Tier-2 approximation error is below the noise floor for all validated modalities at their standard operating SNR. For example, CASSI at standard photon levels has a noise-induced reconstruction error of $\sim 3\%$; the 1% Tier-2 approximation error is therefore dominated by the noise contribution.

Complexity bounds. $N_{\text{max}} = 20$ nodes, $D_{\text{max}} = 10$ depth. These bounds are conservative: no validated modality exceeds 6 nodes or depth 5 (Table 1).

S2. Adjoint Consistency Validation

Every primitive in $\mathcal{B}$ must satisfy the adjoint consistency condition (Equation (3)):

$$\frac{|\langle A\mathbf{x}, \mathbf{y} \rangle - \langle \mathbf{x}, A^\dagger \mathbf{y} \rangle|}{\max(|\langle A\mathbf{x}, \mathbf{y} \rangle|, \epsilon)} < 10^{-6}, \quad (\text{S3})$$

where $\mathbf{x}$ and $\mathbf{y}$ are random vectors and $\epsilon = 10^{-8}$.

Table S1 reports the adjoint consistency check for all 10 primitives.

Table S1: Adjoint consistency validation for all 10 canonical primitives.

Primitive	Relative error	Pass ($< 10^{-6}$)?
Propagate P	$< 10^{-14}$	Y
Modulate M	$< 10^{-15}$	Y
Project Π	$< 10^{-12}$	Y
Encode F	$< 10^{-14}$	Y
Convolve C	$< 10^{-14}$	Y
Accumulate Σ	$< 10^{-15}$	Y
Detect D (linearized)	$< 10^{-13}$	Y
Sample S	$< 10^{-15}$	Y
Disperse W	$< 10^{-13}$	Y
Scatter R	$< 10^{-11}$	Y

S3. Per-Phase Error Bounds

The total approximation error (Equation 15 in the main text) decomposes into per-phase contributions. Here we provide explicit bounds for each physics-stage family.

Propagation ($\varepsilon_{\text{prop}}$). The paraxial (Fresnel) approximation of free-space propagation introduces an error bounded by:

$$\varepsilon_{\text{prop}} \leq \frac{\pi}{4} \frac{a^4}{\lambda d^3}, \quad (\text{S4})$$

where a is the aperture half-width, λ is the wavelength, and d is the propagation distance. For typical imaging geometries ($a \sim 1$ cm, $\lambda \sim 500$ nm, $d \sim 10$ cm), $\varepsilon_{\text{prop}} < 10^{-8}$, far below $\varepsilon = 0.01$.

The evanescent wave truncation error (neglecting spatial frequencies $f > 1/\lambda$) is bounded by the fraction of signal energy above the diffraction limit, which is negligible for band-limited objects.

Elastic interaction (ε_{int}). The Modulate primitive provides exact representation: $\varepsilon_{\text{int}} = 0$. This is because elastic forward interaction at Tier-2 fidelity is, by definition, element-wise multiplication.

Scattering ($\varepsilon_{\text{scat}}$). For single-scattering media (first Born approximation), the Scatter primitive is exact within the Born model: $\varepsilon_{\text{scat}}^{(1)} = 0$. The error relative to the true physics is the Born approximation error itself, which is bounded by $\varepsilon_{\text{scat}} \leq \|V\|/k$, where V is the scattering potential and k is the wavenumber. For Tier-2 media (weak scatterers), this is below ε .

For multiple-scattering media represented by finite Born series (DOT, thick tissue), the L -th order Born approximation has error $\varepsilon_{\text{scat}}^{(L)} \leq (\|V\|/k)^{L+1}$, which converges geometrically.

Encoding–projection (ε_{enc}). Exact: $\varepsilon_{\text{enc}} = 0$. The Radon transform and Fourier encoding are exact linear operators.

Detection–readout (ε_{det}). The detection chain error is dominated by the detector PSF approximation (modeling the spatially varying PSF as shift-invariant within the detector tile). For modern CCD/CMOS detectors with pixel pitch p and inter-pixel crosstalk length ℓ_c :

$$\varepsilon_{\text{det}} \leq \frac{\ell_c^2}{p^2}, \quad (\text{S5})$$

which is $< 10^{-3}$ for typical detectors ($\ell_c \sim 0.1p$).

S4. Complete Modality Decomposition Details

This section provides the detailed DAG decomposition and physics-stage classification for representative modalities.

CASSI (Coded Aperture Snapshot Spectral Imaging).

- **Physics stages:** Source $\rightarrow$ Interaction (coded mask, elastic) $\rightarrow$ Detection (spectral dispersion, spatial integration, photon counting).
- **DAG:** Source $\rightarrow M(\mathbf{m}_{\text{mask}}) \rightarrow W(\alpha, a) \rightarrow \Sigma_\lambda \rightarrow D(g, \eta_{\text{Poisson}})$.
- **Stage-to-primitive mapping:** Interaction $\rightarrow M$; Dispersion $\rightarrow W$; Integration $\rightarrow \Sigma$; Detection $\rightarrow D$.
- $e_{\text{Tier2}} < 10^{-4}$ (the CASSI forward model is exactly linear at Tier-2).

MRI (Magnetic Resonance Imaging).

- **Physics stages:** Source $\rightarrow$ Interaction (coil sensitivity, elastic) $\rightarrow$ Encoding (Fourier, k -space) $\rightarrow$ Detection (undersampling, Gaussian noise).
- **DAG:** Source $\rightarrow M(\mathbf{s}_{\text{coil}}) \rightarrow F(\mathbf{k}) \rightarrow S(\Omega) \rightarrow D(g, \eta_{\text{linear}})$.
- **Stage-to-primitive mapping:** Interaction $\rightarrow M$; Encoding $\rightarrow F$; Sampling $\rightarrow S$; Detection $\rightarrow D$.
- $e_{\text{Tier2}} < 10^{-6}$ (Fourier encoding is exact; coil sensitivity is shift-variant but exactly representable by M).

Compton Scatter Imaging.

- **Physics stages:** Source $\rightarrow$ Interaction (electron density, elastic) $\rightarrow$ Scattering (Klein–Nishina, inelastic) $\rightarrow$ Detection (energy-resolving).
- **DAG:** Source $\rightarrow M(n_e) \rightarrow R(\sigma_{\text{KN}}, \Delta\varepsilon) \rightarrow D(g, \eta_{\text{linear}})$.
- **Stage-to-primitive mapping:** Elastic interaction $\rightarrow M$; Scattering $\rightarrow R$; Detection $\rightarrow D$.

- $e_{\text{Tier2}} < 0.01$ with R ; 0.34 without R (direction change and energy shift cannot be absorbed by other primitives).

Quantum Ghost Imaging.

- **Physics stages:** Source (entangled photon pairs) $\rightarrow$ Interaction (object transmission, elastic) $\rightarrow$ Detection (bucket, photon counting).
- **DAG:** Source $\rightarrow M(\mathbf{m}_{\text{corr}}) \rightarrow \Sigma \rightarrow D(g, \eta_{\text{Poisson}})$.
- **Key insight:** Operator-equivalent to SPC at Tier-2 abstraction. The quantum correlations determine the measurement patterns $\mathbf{m}_{\text{corr}}$, but the forward operator structure is identical to classical single-pixel imaging. The “quantum” aspect resides in the source statistics, not in the operator.
- $e_{\text{Tier2}} < 10^{-4}$.

THz Time-Domain Spectroscopy.

- **Physics stages:** Source (broadband THz pulse) $\rightarrow$ Interaction (sample transfer function, convolution) $\rightarrow$ Detection (coherent field, electro-optic sampling).
- **DAG:** Source $\rightarrow C(\mathbf{h}_{\text{sample}}) \rightarrow D(g, \eta_{\text{coherent}})$.
- **Key insight:** Uses the coherent-field Detect family (η_5 : $g \cdot \text{Re}[\mathbf{x} \cdot e^{i\phi}]$), which measures the electric field rather than intensity. No new primitive required.
- $e_{\text{Tier2}} < 10^{-3}$.

S5. Detect Response Family Analysis

The five canonical Detect response families are designed to cover the physical detection mechanisms in imaging without forming a universal approximator. Here we justify this choice.

Coverage.

- **Families 1–3** (linear-intensity, logarithmic, sigmoid) cover all classical intensity detectors: CCD, CMOS, photomultipliers, scintillators, bolometers. The logarithmic and sigmoid families handle the nonlinear response of wide-dynamic-range and saturating detectors, respectively.
- **Family 4** (Poisson-rate) covers photon-counting detectors and any detector where the dominant noise source is shot noise.
- **Family 5** (coherent-field) covers all interferometric detection systems: heterodyne/homodyne detection (THz-TDS), balanced detection (OCT), and digital holography.

Non-universality. Each family has at most 2 free scalar parameters (g and x_0 or ϕ). The total parameter space is thus at most $\mathbb{R}^2$ per pixel. A universal approximator would require $O(n)$ free parameters to represent an arbitrary function on n inputs. With 2 parameters, the Detect families can represent only a 2-dimensional manifold in the space of all possible detector response functions, confirming that Detect is not a universal approximator.

A modality whose detection mechanism lies outside all five families—for example, a detector with a non-monotonic response curve, or a quantum non-demolition measurement—would signal the need for a sixth family or a new primitive, triggering the extension protocol.

S6. Proof Details: Sub-Multiplicativity Error Bound

We provide the detailed derivation of the total error bound (Equation 15).

Let $H = H_K \circ \dots \circ H_1$ and $H_G = \tilde{H}_K \circ \dots \circ \tilde{H}_1$, where $\tilde{H}_k$ is the primitive realization of H_k with $\|H_k - \tilde{H}_k\| \leq \varepsilon_k$.

By telescoping:

$$H - H_G = \sum_{k=1}^K \tilde{H}_K \circ \dots \circ \tilde{H}_{k+1} \circ (H_k - \tilde{H}_k) \circ H_{k-1} \circ \dots \circ H_1. \quad (\text{S6})$$

Taking operator norms:

$$\|H - H_G\| \leq \sum_{k=1}^K \left(\prod_{j=k+1}^K \|\tilde{H}_j\| \right) \cdot \varepsilon_k \cdot \left(\prod_{j=1}^{k-1} \|H_j\| \right) \quad (\text{S7})$$

$$\leq \sum_{k=1}^K \varepsilon_k \cdot B^{K-1} \quad (\text{since } \|H_j\|, \|\tilde{H}_j\| \leq B) \quad (\text{S8})$$

$$\leq K \cdot \max_k(\varepsilon_k) \cdot B^{K-1}. \quad (\text{S9})$$

For the relative error:

$$\frac{\|H - H_G\|}{\|H\|} \leq \frac{K \cdot \max_k(\varepsilon_k) \cdot B^{K-1}}{\|H\|}. \quad (\text{S10})$$

Since $\|H\| \geq B^{-K+1} \cdot \sigma_{\min}(H)$ (where $\sigma_{\min}$ is the minimum singular value), and for well-conditioned forward models $\sigma_{\min}(H) > 0$, the relative error is bounded by ε provided:

$$\max_k(\varepsilon_k) \leq \frac{\varepsilon \cdot \|H\|}{K \cdot B^{K-1}} \leq \frac{\varepsilon}{K \cdot \kappa(H)}, \quad (\text{S11})$$

where $\kappa(H) = \|H\| \cdot B^{K-1} / \sigma_{\min}(H)$ is the effective condition number of the factorization. For Tier-2 forward models with standard imaging geometries, the per-factor Tier-2 truncation errors satisfy this bound.